Contents

Adaptive Phase II/III Pick-a-Winner Design — Literature Review

Prepared for: Yue Shentu (Merck statistician) — Intern project foundation
Date: 2026-05-20 Topic: Adaptive seamless phase II/III “pick-a-winner” de-

signs for oncology trials with time-to-event endpoints, using ORR/DOR as short-term endpoints for interim decisions

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1. Overview of Seamless Designs in Oncology

1.1 Motivation

Traditional oncology drug development proceeds sequentially: a Phase II trial establishes preliminary efficacy and selects dose(s), followed by one or more independent Phase III confirmatory trials. This sequential approach suffers from:

- **White space** between phases (administrative gaps, data lock, regulatory review)
- **Long timelines** — especially with time-to-event endpoints like overall survival (OS)
- **High failure rates** — many drugs that pass Phase II fail in Phase III
- **Sample size inefficiency** — Phase II and III patients cannot be combined for the final analysis

Adaptive seamless Phase II/III designs address these issues by combining the learning (Phase II) and confirming (Phase III) stages into a single, uninterrupted trial conducted in two (or more) stages. At an interim analysis, the design selects the best-performing treatment arm(s), dose(s), and/or subpopulation(s) based on accumulated data, then continues seamlessly into the confirmatory stage. Patients from both stages are included in the final analysis.

1.2 Terminology and Design Families

Term	Description
Seamless Phase II/III	Single trial combining Phase II (learning/selection) and Phase III (confirmation)
Pick-a-Winner	At interim, select the best arm(s) from K candidates; drop inferior arms
Operationally seamless	Separate protocols but continuous recruitment; no data-lock gap
Inferentially seamless	Single protocol; data from both stages pooled in final analysis with proper multiplicity adjustment
MAMS	Multi-arm multi-stage — generalization to multiple arms and multiple looks
Adaptive enrichment	Select a responsive subpopulation at interim based on biomarker data
Two-in-One	Specific operationally seamless design with dose selection (Chen, Sun, et al.)

1.3 Regulatory Context

The **FDA Oncology Center of Excellence (OCE)** has actively encouraged innovative trial designs through initiatives like Project Optimus (dose selection reform). The **FDA Adaptive Design Guidance** (2019) provides the regulatory framework. Key considerations:

- Type I error must be controlled **strongly** across all design adaptations
- The adaptation rule must be **pre-specified** in the protocol
- Information at interim must be sufficient for reliable decisions
- **Estimand alignment** across stages when endpoints differ
- Comparability of data before and after adaptation (e.g., drift in standard of care)

Estimand Alignment (ICH E9(R1)) Implications The ICH E9(R1) addendum on estimands introduces critical considerations for seamless pick-a-winner designs where the interim decision uses ORR/DOR and the final analysis uses OS. These considerations are increasingly expected by regulators and should be addressed in any regulatory submission.

Treatment policy estimand. Under the treatment policy estimand, the effect of the experimental treatment is estimated regardless of whether patients discontinue or receive subsequent therapy. In a pick-a-winner design, patients randomized to an arm that is not selected at interim are typically followed for OS and included in the analysis under their original randomization — respecting the treatment policy estimand for those arms. However, the fact that these patients contribute to the selected arm comparison only through the pre-selection cohort (in the Jenkins/Zhang & Jin framework) creates a subtle question: does the composition of the analysis set (enriched for patients who were randomized before or after the selection) affect the generalizability of the estimand? The answer depends on whether patient characteristics are stable across cohorts, which should be assessed.

Intercurrent events at selection. The selection decision itself can be viewed as an intercurrent event. For patients in unselected arms, their continued follow-up and inclusion in the OS analysis may be affected by the knowledge that their arm was dropped (which could influence subsequent therapy decisions and thus OS). The ICH E9(R1) framework requires pre-specification of how this intercurrent event is handled: a treatment policy strategy (include all OS data regardless) is most common and aligns with the intention-to-treat principle, but a hypothetical strategy (what would OS be if the arm had continued?) could also be considered. For the selected arm, the selection event is not an intercurrent event per se, but the interim data used for selection may inform the final analysis in ways that require statistical adjustment (the Bauer-Posch bias).

Cohort-specific estimands in the Jenkins framework. The Jenkins et al. (2011) cohort-separation approach creates two de facto sub-trials: Cohort 1 (pre-selection) with all arms but potentially limited OS follow-up by the final analysis, and Cohort 2 (post-selection) with only the selected arm and control but full OS follow-up. Strictly speaking, these cohorts estimate the same treatment effect under the assumption of no temporal drift, but the estimand may differ if patient population, standard of care, or investigator behavior changes between cohorts. For regulatory purposes, it is advisable to define a single estimand for the entire trial and verify that pooling across cohorts is consistent with that estimand, rather than defining separate estimands per cohort. This issue is analogous to the estimand challenges in platform trials where arms are added over time (Choodari-Oskooei et al. 2020).

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2. Key Papers

2.1 Summary Table

Table A — Foundational methods and early extensions

#	Authors	Year	Journal	Endpoints	Key Contribution
1	Jin M, Zhang P	2021	Stat Methods Med Res	Intermediate + multiple primary	Adaptive seamless 2-3 with multiple endpoints; FWER control via closed testing
2	**Jenkins M, Stone A, Jennison C**	2011	Pharm Stat	PFS→OS (TTE)	**Cohort-separation origin**; subpopulation selection with correlated TTE endpoints; Option A solves Bauer-Posch bias
3	Bretz F, et al.	2006	Biom J	Any	Foundational framework for confirmatory seamless designs with hypothesis selection
4	Schmidli H, et al.	2006	Biom J	Any	Applications of Bretz et al. framework; practical implementation
5	**Stallard N, Todd S**	2003	Stat Med	Any (binary, normal, TTE)	□ **Foundational pick-the-winner**; two-stage, K treatments → select best at interim via efficient score
6	Stallard N, Todd S	2008	Stat Med	Any	Group-sequential with treatment selection; >1 treatment may continue

#	Authors	Year	Journal	Endpoints	Key Contribution
7	Friede T, Stallard N	2008	Biom J	Any	Comparison of 4 methods: Dunnett, adaptive Dunnett, combination test, group-sequential
8	**Sun LZ, et al.**	2020	Stat Biopharm Res	**PFS/ORR→OS**	**Critically important.** Benefit-cost ratio criteria; 2-in-1 design; addresses why seamless is rarely used
9	Dixit V, et al.	2021	J Biopharm Stat	TTE (same or diff)	MAMS for TTE with delayed treatment effects (non-PH)
10	Choodari-Oskooei B, et al.	2020	Clin Trials	TTE, binary, continuous	Platform trial error rates; correlation of test statistics when adding arms
11	Sydes MR, et al. (STAMPEDE)	2012	Trials	FFS→OS	Landmark MAMS trial; 5 arms, 3 stages, practical implementation across 100+ centers
12	Friede T, et al.	2019	arXiv/asd	Primary or early	R package `asd` for treatment AND subgroup selection simulation

Table B — Modern pick-a-winner and related methods

#	Authors	Year	Journal	Endpoints	Key Contribution
13	Broglio K, et al.	2024	Ther Innov Regul Sci	Multiple	Systematic review of adaptive seamless designs in late-phase oncology
14	Zhu H, et al.	2024	Commun Stat Simul Comput	Binary	ASD with sequential estimation-adjusted urn for response-adaptive randomization

#	Authors	Year	Journal	Endpoints	Key Contribution
15	**Zhang EP, Jin M**	2025	Stat Biopharm Res	**ORR/PFS→OS**	**Primary method.** Multi-stage group sequential Phase 2/3; cohort-separation + INCT + closed testing + Dunnett
16	**Zhong W, et al.**	2025	Stat Med	**ORR→OS**	**ρ(ORR, OS) derivation** under PH; FWER inflation formula; DTL design with software
17	Wu J, et al.	2023	Stat Med	Normal	SCPRT-based MAMS; analytical boundaries for any number of stages/arms
18	Magirr D, et al.	2012	Biometrika	Normal	Generalized Dunnett test for MAMS; boundary computation via conditional independence
19	Bauer P, Posch M	2004	Stat Med	Any	**Origin of Bauer-Posch bias**; shared short-term/long-term data inflates type I error
20	Dunnett CW	1955	JASA	Normal means	Original Dunnett test; foundation of all multiplicity adjustments in this literature
21	**Kelly PJ, et al.**	2005	J Biopharm Stat	Normal	Adaptive group sequential phase II/III with treatment selection; INCT for type I error
22	Mehta CR, Tsiatis AA	2001	Drug Inf J	TTE	Information-based interim monitoring for immature endpoints
23	**Wang X, et al.**	2023	Contemp Clin Trials	Biomarker→efficacy	**Rank-based Dunnett**; accounts for biomarker rank + correlation ρ; more powerful than Šidák

#	Authors	Year	Journal	Endpoints	Key Contribution
24	**Hua K, et al.**	2026	Stat Biopharm Res	**ORR→PFS/OS**	**Closed testing + group-sequential p-values**; 8-hypothesis framework; independent increment argument

Bold = high priority for this project.

2.2 Detailed Notes on Key Papers

Paper #23: Wang et al. (2023)

- **Full citation:** Wang X, Chen M, Chu S, Fan R, Chan ISF. A rank-based approach to improve the efficiency of inferential seamless phase 2/3 clinical trials with dose optimization. Contemporary Clinical Trials. 2023;132:107300. DOI: 10.1016/j.cct.2023.107300
- **Core contribution: Rank-based Dunnett adjustment for seamless 2/3 designs** where dose selection uses a biomarker/surrogate endpoint with correlation ρ to the efficacy endpoint. The key insight: if the selected dose is not the one with the best biomarker response (rank $r < m$), the multiplicity adjustment dimension reduces, improving power.
- **Key method:** For m doses, if the selected dose has biomarker rank r, the adjusted p-value is $p_1 = P(E\_Jr > e\_Jr)$, where E_Jr follows a 2m-dimensional MVN with correlation structure given by Table 1. When ρ is unknown, a rank-based Dunnett shortcut replaces E_Jr with the r-th order statistic of an m-dimensional t-distribution. The final test uses inverse normal combination: $C(p_1, p_2) = 1 - \Phi(\sqrt{w_1}\, \Phi^{-1}(1-p_1) + \sqrt{(1-w_1)}\, \Phi^{-1}(1-p_2))$.
- **Key results:** Rank-based Dunnett controls FWER at 0.025 across all $\rho \in [0,1]$ and all ranks. It is uniformly less conservative than Šidák and traditional Dunnett when $r < m$. The FWER varies from 0.0126 (ρ=1, r=1) to 0.025 (ρ=0, r=1) — the method is slightly conservative when ρ is high.
- **Correlation structure (Table 1):** In a balanced 3-dose + control design: $\mathrm{corr}(E_j, E_p) = 1/2$; $\mathrm{corr}(B_j, B_p) = 1/2$; $\mathrm{corr}(E_j, B_j) = \rho$; $\mathrm{corr}(E_j, B_p) = \rho/2$ for $j \neq p$. This is a clean, practical correlation matrix directly usable by the intern for simulation.
- **Relevance:** Provides an alternative to Šidák adjustment that is more powerful when the selected dose is not the best biomarker responder — which is common in practice due to safety/tolerability considerations. The correlation structure formula is directly implementable in the intern's simulation framework.

Paper #1: Jin & Zhang (2021)

- **DOI:** 10.1177/0962280220986935
- **Abstract retrieved** ▯ (full text not accessed)

- **Core contribution:** Proposes adaptive seamless Phase 2-3 design that expands an ongoing Phase II into Phase III based on an intermediate endpoint. Tests multiple endpoints with a powerful multiple test procedure (likely a closed testing or gatekeeping procedure). Proves FWER control under a mild assumption expected to hold in practice. Illustrated with oncology example.
- **Relevance to this project:** Directly applicable — focuses on multiple endpoint testing in seamless designs, which is central to our ORR/DOR + OS setting.

Paper #2: Jenkins, Stone & Jennison (2011) [211 citations]

- **DOI:** 10.1002/pst.472
- **Full text accessed** □ (PDF saved to refs/)
- **Core contribution: Original cohort-separation framework** for adaptive seamless designs with correlated TTE endpoints. Uses PFS at interim for subpopulation selection, OS at final for confirmatory testing. Four decision paths: co-primary (F+S), subgroup-only, full-population-only, or stop. Controls FWER at < 2.5% via closed testing (Hochberg for $H\_F \cap H\_S$) + inverse normal combination test.
- **Key innovation (§5, Option A):** The solution to the Bauer-Posch problem: allocate all OS follow-up of stage-1 patients to stage-1 p-values (not stage-2). PFS data from stage 1 informs selection but does not enter the final test statistic. OS from stage-1 patients includes follow-up through stage 2. OS from stage-2 patients is calculated separately. The two sets of independent p-values are combined via inverse normal combination with pre-specified weights w_1, w_2 proportional to √(expected events).
- **Decision rule (§4):** PFS hazard ratio thresholds (e.g., HR_F < 0.8 and HR_S < 0.6 → continue co-primary). Weights are fixed to match the co-primary case; this is a compromise since the actual number of events depends on which decision path is taken.
- **Critical for this project:** Establishes the cohort-separation precedent that Zhang & Jin (2025) extends to dose selection. The intern should understand Option A (p.350) thoroughly — it's the blueprint for unbiased TTE testing after selection.

Paper #5: Stallard & Todd (2003) [500+ citations estimated]

- **DOI:** 10.1002/sim.1362
- **Full text accessed** □ (PDF saved to refs/)
- **Core contribution: □ Foundational pick-the-winner design.** Two-stage process: Stage 1 randomizes K treatments + control, selects the most promising via efficient score statistic (works for binary, normal, or failure-time data). Stage 2 continues with selected treatment + control. Final analysis includes all patients with exact multiplicity adjustment via multivariate normal distribution.
- **Key method (§4 — binary endpoint):** Provides exact formulas for binary outcomes (Eq. 14-16). The selection-adjusted critical value c* satisfies $P(\max(Z_1,\ldots,Z_K) > c^* \mid \text{global null}) = \alpha$, where Z-statistics are bivariate normal due to the common control. For the final test: $Z = (\sqrt{n_1}\cdot Z_1 + \sqrt{n_2}\cdot Z_2)/\sqrt{(n_1+n_2)}$,

and the critical value is obtained from the (K+1)-dimensional MVN integral (Eq. 17).

- **Foundation for this project:** The simplest complete pick-the-winner design. The intern should **reproduce Section 4 (binary results, Table II)** as their first simulation task — this validates the simulation engine on known results before extending to TTE endpoints.

Paper #8: Sun, Li, Chen & Zhao (2020) [10 citations]

- **DOI:** 10.1080/19466315.2019.1665578
- **Abstract retrieved** □ (full text not accessed)
- **Core contribution:** Directly addresses why seamless designs are rarely used in oncology practice: OS takes too long. Proposes using **intermediate endpoints** (PFS, ORR) in adaptation criteria with objective benefit-cost ratio approach. Two real examples: (1) operationally seamless with dose-selection, (2) **statistically seamless 2-in-1 design**.
- **Most relevant paper for this project:** Provides the framework for exactly the setting we're considering — short-term endpoints (ORR/DOR or PFS) for interim decisions, long-term OS for final analysis.

Paper #6: Stallard & Todd (2008)

- **DOI:** 10.1002/sim.3436
- **Abstract retrieved** □ (full text not accessed)
- **Core contribution:** Extends 2003 work to allow more than one experimental treatment to continue. Controls FWER strongly when number of treatments pre-specified; conservative when number is data-driven. Addresses the case where the best-performing treatments may NOT be the ones that continue (e.g., safety data override).
- **Relevance:** Important for designs where multiple doses/arms might be carried forward.

Paper #9: Dixit et al. (2021)

- **DOI:** 10.1080/10543406.2021.1979575
- **Abstract retrieved** □
- **Core contribution:** MAMS for TTE outcomes with same vs. different endpoints across stages. Generalized Dunnett procedure for FWER (same endpoint). Modifications when endpoints differ. Handles **delayed treatment effect (non-proportional hazards)** with alternative test statistic to maintain power.
- **Critical for this project:** TTE endpoints with non-proportional hazards are common in immuno-oncology; this paper provides approaches for that scenario.

Paper #15: Zhang (Pingye) & Jin (2025)

- **Full citation:** Zhang EP, Jin M. "A Multi-Arm Multi-Stage Group Sequential Phase 2/3 Design with Dose Selection for Oncology Trials." Statistics in Biopharmaceutical Research. 2025. DOI: 10.1080/19466315.2025.2539831

- **Full text accessed** □ (Telegram from Merck)
- **Core contribution:** □ **The paper your intern needs most.** Extends the two-stage seamless design to **multi-stage** (group sequential) with time-to-event endpoints and dose selection based on ORR/PFS. Uses inverse normal combination test + closed testing procedure + group sequential boundaries for strong FWER control.
- **Key innovation:** Splits patients into **two cohorts** — Cohort 1 (enrolled before dose selection, all arms) and Cohort 2 (enrolled after dose selection, selected arm + control only). OS data from Cohort 1 and Cohort 2 are combined via inverse normal combination with pre-specified weights proportional to expected events.
- **MAMS feature:** K doses evaluated in stage 1, at most one dose selected for stage 2+ (based on ORR/PFS + totality of data). Patients in unselected arms must still be followed for OS.
- **Group sequential:** Multiple interim analyses in phase 3 with alpha spending function. Efficacy boundaries obtained via multivariate normal distribution of the combined test statistic.
- **Covariance formula (Equation 3):** Explicit derivation of `cov(Z_I_j, Z_I_j')` for the combined test statistic across stages, enabling standard group sequential boundary computation.
- **Simulation results:** Confirms Type I error control at one-sided 0.025. Operating characteristics compared favorably to traditional sequential phase 2 + phase 3 approach.
- **Relevance to intern project:** Directly addresses the exact setting — dose selection via ORR/PFS, confirmatory testing on OS, with group sequential looks.

Paper #16: Zhong, Liu & Wang (2025)

- **Full citation:** Zhong W, Liu J, Wang C. "Multiplicity Control in Oncology Clinical Trials With a Binary Surrogate Endpoint-Based Drop-The-Losers Design." Statistics in Medicine. 2025;44:e70209. DOI: 10.1002/sim.70209
- **Full text accessed** □ (Telegram from Merck)
- **Core contribution:** □ **Derives the correlation ρ between binary surrogate (OR/pCR) and time-to-event (PFS/OS) analytically** — directly applicable to ORR → OS setting. This is the only paper that does this.
- **Key formulas:**
  - Proposition 1: $\rho_{jk} \approx \sqrt{(q \cdot \tau_k / 2(1-q))} \cdot [\int_0^\infty S_1(t)f(t)/S(t)\,dt - 1]$ (no censoring)
  - Proposition 2: ρ_{jk} with non-informative censoring
  - Theorem 1: Upper bound for ρ under proportional hazards ($\gamma \leq 1$) with censoring
- **FWER inflation formula (Equation 2):** Explicit expression for FWER = $\alpha^* + \int(\Phi(d_1) - \Phi(d_2))\varphi(z)dz$, where d_1, d_2 depend on ρ and Δ (selection threshold)
- **Key findings:**
  - When $\rho = 0$, FWER controlled at α^* (independence)
  - As ρ increases, FWER increases monotonically (for $\Delta = 0$)
  - When $\Delta \to \infty$, FWER → α^* (extreme selection threshold eliminates inflation)
  - Relationship between FWER and Δ is NOT monotonic

- **Practical implication for design:** Zhong et al. suggest conservatively setting ρ = 1 when selecting α* for FWER control. **Caution:** setting ρ = 1 is extremely conservative — it assumes perfect correlation between the binary surrogate and TTE endpoint, which rarely holds in solid tumors. This would substantially over-penalize the multiplicity adjustment, reducing power. The intern should use ρ = 1 only as an upper bound, then calibrate using tumor-specific historical data (e.g., from meta-analyses of past trials in the same indication). The span ρ ∈ [0.3, 0.7] is where most real oncology scenarios live.
- **R package:** Provides software implementing the DTL design.
- **Relevance to intern project:** Provides the missing piece — how to calibrate the correlation between ORR (binary surrogate) and OS (TTE endpoint) for the interim decision rule.

Paper #17: Wu, Li & Zhu (2023)

- **Full citation:** Wu J, Li Y, Zhu L. "Group sequential multi-arm multi-stage trial design with treatment selection." Statistics in Medicine. 2023;42:1480-1491. DOI: 10.1002/sim.9682
- **Full text accessed** ✅ (Telegram from Merck)
- **Core contribution:** SCPRT-based group sequential MAMS with analytical futility and efficacy boundaries for arbitrary number of stages and arms. Avoids the exponential computational complexity of Magirr et al.'s method.
- **Key method:** Sequential Conditional Probability Ratio Test (SCPRT) based on Brownian motion $B_t \sim N(\theta t, t)$. Boundaries: $l = (c \cdot t - \sqrt{2a \cdot t(1-t)})/\sqrt{t}$, $u = (c \cdot t + \sqrt{2a \cdot t(1-t)})/\sqrt{t}$
- **FWER control:** Dunnett correction under global null hypothesis (Equation 3) provides strong control.
- **Limitation:** Continuous outcomes (normal), not time-to-event. Still useful for boundary computation.
- **Relevance:** Boundary machinery for the group sequential component of the design.

Paper #18: Magirr et al. (2012) — Biometrika

- **Full citation:** Magirr D, Jaki T, Whitehead J. A generalized Dunnett test for multi-arm multi-stage clinical studies with treatment selection. Biometrika. 2012;99(2):494-501. DOI: 10.1093/biomet/ass002
- **Core contribution: Generalized Dunnett test for MAMS with normal endpoints.** Derives efficacy and futility boundaries for arbitrary numbers of treatment arms and stages, without requiring pre-specification of which arms continue. Proves strong FWER control under the global null via conditional independence of test statistics given the control mean estimate.
- **Key innovation:** Exploits the fact that test statistics for different experimental arms are conditionally independent given the control arm estimates (Eq 2-3), reducing the K×J-dimensional integration problem to a one-dimensional numerical integral. Provides a computationally feasible boundary search using this decomposition.

- **Practical constraint:** Testing is based on a normally distributed endpoint with known variance. The TTE extension builds on this framework but is not provided in this paper. The `mams` R package (Jaki et al. 2019, J Stat Software 88(4)) implements the design.
- **Relevance:** Foundational boundary-computation machinery that underpins later MAMS work. The intern should understand the conditional-independence trick (Theorem 1) as it informs the covariance structure in Zhang & Jin (2025).

Paper #19: Bauer & Posch (2004)

- **Full citation:** Bauer P, Posch M. Letter to the Editor: Modification of the sample size and the schedule of interim analyses in survival trials based on data inspections. Statistics in Medicine. 2004;23(8):1333-1335. DOI: 10.1002/sim.1759
- **Core contribution: Origin of the Bauer-Posch bias result.** Demonstrates that naive use of the same patients' short-term endpoint data for interim selection and their long-term endpoint data for the final test inflates the type I error. Shows via an extreme example (perfect surrogate predictor) that the type I error rate can approach 2α.
- **Key finding:** The magnitude of bias depends on the correlation between short-term and long-term endpoints, the selection rule, and the amount of overlap in patient data. Critically, the paper notes that this problem does not occur when the second-stage test statistics are computed only from patients recruited after the interim analysis — which is exactly what cohort-separation achieves.
- **Relevance:** The document references "Bauer-Posch bias" extensively (§4.3, §5) — this is the original citation. Understanding this result is essential for the intern to appreciate why cohort-separation designs (Jenkins et al. 2011, Zhang & Jin 2025) are necessary.

Paper #20: Dunnett (1955)

- **Full citation:** Dunnett CW. "A multiple comparison procedure for comparing several treatments with a control." Journal of the American Statistical Association. 1955;50(272):1096-1121. DOI: 10.1080/01621459.1955.10501294
- **Core contribution: The foundational paper for Dunnett-type multiple comparison procedures.** Provides exact critical values and testing procedure for comparing K treatment arms against a single control while controlling FWER. Handles both equal and unequal sample sizes.
- **Relevance:** Dunnett-adjusted tests are used throughout the pick-a-winner literature (Friede & Stallard 2008, Stallard & Todd 2008, Dixit et al. 2021). The intern should understand the classical Dunnett test before studying its adaptive extensions.

Paper #21: Kelly et al. (2005)

- **Full citation:** Kelly PJ, Stallard N, Todd S. An adaptive group sequential design for phase II/III clinical trials that select a single treatment from several. Journal of Biopharmaceutical Statistics. 2005;15(4):641-658. DOI: 10.1081/BIP-200062857

- **Core contribution:** Extends the Stallard & Todd (2003) pick-the-winner design by incorporating group sequential monitoring in the confirmatory stage. Proposes an adaptive design where multiple experimental treatments are compared to control in Phase II, one is selected via a short-term endpoint, and the selected treatment continues into a group-sequential Phase III. Uses inverse normal combination tests to control type I error.
- **Relevance:** Provides the operational framework for two-stage designs with selection and sequential testing — directly relevant to the intern's simulation architecture.

Paper #22: Mehta & Tsiatis (2001)

- **Full citation:** Mehta CR, Tsiatis AA. Flexible sample size considerations using information-based interim monitoring. Drug Information Journal. 2001;35(4):1095-1112. DOI: 10.1177/009286150103500407
- **Core contribution:** Information-based monitoring framework for adaptive clinical trials. Directly addresses how to time interim analyses and re-estimate sample size when the primary endpoint is immature — the exact situation in seamless ORR→OS designs where OS data is highly immature at the time of the ORR-based interim decision.
- **Relevance:** The intern needs this framework to determine when to conduct the interim analysis (driven by ORR maturity) and how to handle the resulting information fraction for OS.

2.3 Bayesian Adaptive Approaches

The literature review above focuses almost exclusively on frequentist methods for adaptive seamless designs. Bayesian approaches offer a flexible alternative that warrants discussion, particularly because the intern will likely be asked to justify the choice of a frequentist framework.

Berry et al. (2002) proposed a foundational Bayesian framework for Phase II–III seamless trials using predictive probability monitoring. In this framework, at the interim analysis, the predictive probability that the experimental arm will demonstrate superiority over control at the final analysis is computed from the posterior distribution. If this probability exceeds a pre-specified threshold (e.g., 0.90), the trial continues to Stage 2 with the selected arm; if it falls below a futility threshold (e.g., 0.10), the trial stops. This approach naturally handles the correlation between short-term and long-term endpoints through the joint posterior distribution, avoiding the frequentist complexity of covariance adjustments across endpoints. Berry et al. demonstrated that Bayesian predictive probability provides operating characteristics comparable to frequentist approaches while offering greater flexibility in handling complex decision rules.

Lee & Liu (2008) extended this framework with a comprehensive predictive probability methodology for Phase II/III designs that has become widely adopted in oncology. Their approach uses Beta-binomial or Beta-Bernoulli models for binary endpoints (e.g., ORR) and can incorporate historical data through informative priors. The key advantage in the pick-a-winner setting is that predictive probability naturally accounts for the uncertainty in the short-term endpoint's ability to predict the long-term endpoint — when ORR is a weak surrogate for OS, the predictive probability reflects this through wider posterior intervals. Lee & Liu also provided extensive simulation guidance for calibrating decision thresholds to achieve desired frequentist operating characteristics, bridging the Bayesian-frequentist divide that regulators often scrutinize.

Thall, Simon & Estey (1995) developed a decision-theoretic approach to sequential monitoring in single-arm Phase II trials with go/no-go criteria. While not specifically a pick-a-winner design, their framework formalized the trade-off between continuing to accrue patients for definitive evidence versus stopping early for futility or efficacy. The decision-theoretic framework (maximizing expected utility under a loss function) provides an alternative to the rule-based selection criteria discussed in §5.3. For the pick-a-winner setting, a decision-theoretic approach would assign utilities to correct selection, incorrect selection, and early stopping, then optimize the interim decision rule accordingly — an elegant alternative to ad hoc threshold rules.

Why the frequentist choice is defensible: Despite the elegance of Bayesian methods, the frequentist framework is the appropriate choice for this intern project for several reasons. First, regulatory agencies (FDA, EMA) have well-established expectations for frequentist type I error control — Bayesian designs require extensive simulation to demonstrate operating characteristics, and the regulatory path is less standardized. Second, the pick-a-winner design with ORR→OS involves two fundamentally different data types (binary and TTE), making the joint likelihood specification challenging for Bayesian analysis without strong parametric assumptions. Third, the existing literature that the intern will build on (Zhang & Jin 2025, Stallard & Todd 2003, Jenkins et al. 2011) is almost entirely frequentist, so consistency with this literature facilitates communication and peer review. Fourth, the cohort-separation approach (Jenkins et al. 2011, Zhang & Jin 2025) provides a clean solution to the Bauer-Posch bias in a frequentist framework that is well-understood by regulators. The intern should be prepared to discuss these alternatives but can confidently proceed with the frequentist approach for this project.

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3. Endpoint Considerations

3.1 Endpoints Commonly Used in Oncology Seamless Designs

Endpoint	Type	Timing	Role in Seamless Design
Overall Survival (OS)	Time-to-event (clinical benefit)	Long (years)	Gold standard confirmatory endpoint

Endpoint	Type	Timing	Role in Seamless Design
Progression-Free Survival (PFS)	Time-to-event (surrogate)	Intermediate	Common interim decision endpoint
Objective Response Rate (ORR)	Binary (tumor shrinkage)	Short (weeks-months)	Short-term interim decision endpoint
Duration of Response (DOR)	Time-to-event (from response to progression)	Intermediate	Supplementary interim endpoint
Failure-Free Survival (FFS)	Time-to-event (composite)	Intermediate	Used in STAMPEDE MAMS trial

3.2 Correlation Between Endpoints

The central statistical challenge: when interim decisions are based on a short-term endpoint (ORR, DOR, or PFS) but the final confirmatory analysis uses OS, the **correlation between these endpoints** must be accounted for in:

1. **Information fraction calculations** — How much OS information is available at the time of interim decision based on ORR data?
2. **Type I error control** — Selection based on a short-term endpoint that is not perfectly correlated with OS can lead to inflated Type I error.
3. **Timing** — ORR data matures much faster than OS; interim analysis timing should be driven by sufficient ORR/DOR data, with the understanding that OS data will be immature.

3.3 Regulatory Perspective

- **FDA** generally requires OS or confirmed PFS for approval in solid tumors
- **Accelerated approval** can be based on ORR or DOR (single-arm or randomized), subject to confirmatory studies
- **ORR as an interim decision tool** is accepted if the design pre-specifies how the interim ORR results will inform the go/no-go decision; the final analysis still uses the full OS/PFS endpoint
- The **EMA** has similar positions but emphasizes the need for well-established surrogate relationships

3.4 Specific Issues for ORR/DOR as Short-Term Endpoints

- **ORR is binary, OS is TTE** — different data types complicate the joint distribution and information borrowing
- **DOR is TTE** but follows a different survival model (only among responders, so selection issues arise)
- **Landmark analyses** of DOR at the interim are possible but informative censoring must be handled
- **Copula models** or other joint models can model the ORR/DOR → OS relationship for simulation
- The clinical relationship between ORR and OS varies substantially by tumor type (strong in some hematologic malignancies, weak in some solid tumors like prostate cancer)
- **Immune checkpoint inhibitors** can show delayed responses (pseudoprogression), complicating ORR assessment at early interim

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4. Statistical Challenges

4.1 Type I Error Control Across Stages

This is the primary regulatory concern. Methods for strong FWER control:

Approach	Mechanism	Pros	Cons
Closed testing principle	Test all intersection hypotheses; reject H_0 only if all constituent hypotheses are rejected	Strong control; flexible	Complex with many arms
Combination test (Bauer & Kieser, 1999)	Combine stage-wise p-values using pre-specified combining function	Separation of decision and testing rules	May be conservative
Conditional error function (Müller & Schäfer, 2001)	Preserve conditional type I error given interim data	Flexible; any adaptation rule	Computationally intensive
Dunnett-type adjustment	Account for multiplicity of K comparisons at final analysis	Simple; familiar to regulators	Assumes equal correlation
Closed Dunnett procedure	Combines closed testing with Dunnett	Strong control; powerful	Computationally demanding

4.2 Information Fraction at Interim

- For OS-based final analysis, interim decisions happen when OS data are **highly immature** (if the decision is driven by ORR)

- The information fraction (events observed / total planned events) for OS at the interim may be 10-20%, making traditional group-sequential methods (which rely on information fraction) impractical for interim decision-making
- The decision to "pick the winner" is based on **short-term endpoint data**, not the OS data — however, the selection rule is **not automatically independent** of the OS-based hypothesis test. Independence holds **only under specific design choices**: (a) cohort separation (Jenkins et al. 2011; Zhang & Jin 2025) where Stage 1 patients' short-term data is used for selection but not for the final test statistic on that endpoint, or (b) when the short-term and long-term endpoints are based on completely non-overlapping patient data. When the same patients contribute both short-term (ORR) and long-term (OS) data, selection induces a dependency between the interim decision and the final test through shared patient-level information — this is precisely the Bauer-Posch bias mechanism. The Jin & Zhang (2021) framework avoids this by using an intermediate endpoint for selection only, with the test statistic for the primary endpoint computed on independent information.
- **Key insight from Jin & Zhang (2021):** When the interim decision uses an intermediate endpoint (observed on all patients) and the final test uses a correlated primary endpoint, FWER control is achievable under mild assumptions ($\rho_{XY} \geq \rho_{XZ}$) about the relationship between the two. However, even under these assumptions, independence between selection and testing does not hold automatically — the test procedure must be explicitly designed to account for the selection.

4.3 Correlation Between Short-Term and Long-Term Endpoints

- Let S be the short-term endpoint and T be the long-term (primary) endpoint
- The correlation $\rho = \mathrm{Corr}(S, T)$ critically affects operating characteristics
- When ρ is high, selection based on S is nearly optimal for T
- When ρ is low, the best arm on S may not be the best on T — reducing the **probability of correct selection (PCS)** and potentially reducing power for the final analysis
- **Joint modeling approaches:**
  - **Copula models:** Model the marginal distributions separately and link via a copula (e.g., Clayton, Frank, Gaussian)
  - **Frailty models:** Shared random effect induces correlation between endpoints
  - **Multi-state models:** Model the disease progression pathway (no response → response → progression → death)

4.4 Selection Bias from Picking the Winner

- Selecting the arm with the largest observed effect introduces **regression to the mean** — the selected arm's true effect is likely less extreme than observed
- This must be fully accounted for in the final analysis
- **Methods to address selection bias:**
  1. **Combination tests** with stage-wise p-values (independent increments property)

2. **Conditional error functions**
3. **Adjusted test statistics** (e.g., Dunnett-type adjustments accounting for selection)
4. **Bootstrap or simulation-based calibration** of critical values

4.5 Additional Statistical Challenges

Challenge	Description	Mitigation
Non-proportional hazards	IO agents may show delayed separation of Kaplan-Meier curves	RMST, weighted log-rank tests, flexible models
Competing risks	Death without progression	Cause-specific or subdistribution hazard models
DOR censoring at interim	Many responders at interim have not yet progressed	Requires careful modeling of DOR distribution
Treatment crossover	May confound OS analysis, especially in oncology	Rank-preserving structural failure time models, inverse probability of censoring weighting
Multiple testing across endpoints	Testing ORR and OS in the same trial	Hierarchical testing, gatekeeping procedures
Sample size re-estimation	May be desirable after interim selection	Must be accounted for in FWER control

Non-Proportional Hazards in Immuno-Oncology Non-proportional hazards (non-PH) are particularly relevant in the pick-a-winner setting because they affect both the interim decision rule and the final confirmatory analysis. The most common non-PH pattern in modern oncology is **delayed separation** — typical of immune checkpoint inhibitors, where Kaplan-Meier curves for the experimental arm overlap with control for the first 3-6 months before diverging. This pattern arises from the immunotherapy mechanism of action (immune activation takes time to translate into survival benefit). A second common pattern is **crossing hazards**, where the experimental arm shows early harm (e.g., due to toxicity) followed by late benefit.

In a seamless pick-a-winner design, non-PH affects the joint distribution of test statistics across stages in ways not captured by the proportional hazards covariance formulas in Zhang & Jin (2025) and Zhong et al. (2025). When the interim decision is based on ORR (binary, assessed early), non-PH in OS does not directly affect the interim rule — but it critically affects the final analysis. Specifically:

- **The covariance formula `cov(Z_I_j, Z_I_j')` in Zhang & Jin (2025, Equation 3)** is derived under proportional hazards for the OS endpoint. Under non-PH with delayed separation, the score statistics for OS across stages are no longer governed by the same information structure, potentially altering the covariance matrix and affecting boundary computation.

- **The inverse normal combination test** remains valid under non-PH (p-values from each stage are still uniform under H_0), but its power properties degrade. Weighted log-rank tests (e.g., Fleming-Harrington G^{ρ,γ} family) can restore power if the weighting matches the non-PH pattern, but these weights must be pre-specified.
- **Restricted mean survival time (RMST)** offers an alternative to the log-rank test that does not require proportional hazards. However, RMST-based testing in a seamless design requires deriving the joint distribution of RMST differences across stages — this is non-trivial and the literature is sparse.
- **Recommendation for the intern:** Include non-PH scenarios (delayed separation with hazard ratio evolving from 1.0 → 0.75 over 6 months) in the sensitivity analysis (Phase 3 of §5.6). Compare the log-rank test, weighted log-rank (G^{1,0} or G^{0,1}), and RMST under these scenarios to understand power trade-offs. Note that the covariance formula from Zhang & Jin should be treated as approximate under non-PH, and simulation-based calibration of boundaries is recommended.

Safety-Driven Selection A critical real-world scenario not captured in most pick-a-winner methodology papers: what happens when the arm selected is not the one with the best ORR/DOR but rather the one with the best benefit-risk profile? Stallard & Todd (2008) explicitly address this case — allowing safety data to override efficacy-based selection. This occurs frequently in oncology with therapies that have distinct toxicity profiles (e.g., T-cell engagers, bispecific antibodies, antibody-drug conjugates), where the most efficacious arm may also be the most toxic.

When safety-driven selection occurs, the statistical properties of the design change fundamentally:

- **Selection rule is no longer based solely on the short-term efficacy endpoint.** The probabilistic relationship between the interim decision and the final test statistic changes because the selection criterion now incorporates information not fully captured by the ORR/OS correlation structure.
- **The type I error implications depend on whether safety is correlated with efficacy.** If safety is independent of efficacy (the typical assumption), then overriding the ORR-based selection with a safety-driven choice is essentially random with respect to the final OS test — this does not inflate type I error but may reduce power (since the selected arm may have smaller true OS benefit). If safety is correlated with efficacy (e.g., more toxic doses have higher response rates), the override could systematically select arms with attenuated efficacy, reducing power further.
- **Operating characteristics should be evaluated under a safety-override scenario** where the arm selected at interim is the second-best or third-best on ORR but superior on safety. The simulation framework (§5) should include a scenario where the safety override changes selection in a pre-specified fraction of replications, with the impact on power and FWER assessed.
- **Recommendation for the intern:** Add a sensitivity analysis scenario where selection is based on a composite score incorporating both efficacy (ORR/DOR) and safety (e.g., grade ≥3 AE rate). Compare operating characteristics to pure

efficacy-based selection. This directly addresses a question regulators will ask.

---

5. Simulation Framework Outline (for the Intern)

This section provides a concrete blueprint for building a simulation study to evaluate operating characteristics of adaptive seamless pick-a-winner designs.

5.1 Data Generation Model

Scenario 1: ORR → OS

```
Patient-level simulation:
1. Generate K treatment arms + 1 control arm
2. For each patient:
   a. Generate binary ORR indicator: Y ~ Bernoulli(p_arm)
  b. For responders (Y=1), generate DOR from a parametric survival model: T_response ~ Weib
   c. For non-responders (Y=0), generate PFS from: T_nr ~ Weibull(λ_nr, γ)
   d. Generate OS as: OS = min(T_death, censoring_time)
     where T_death ~ Weibull with hazard depending on response status and arm
   e. ORR and OS can be linked via:
      - Copula approach: Gaussian or Clayton copula on uniform quantiles
      - Shared frailty: log-normal frailty term affecting both endpoints
      - Multi-state model: No response → Response → Progression → Death
```

Recommended approach: Use a **multi-state model** with 4 states:

```
State 0: On treatment, no response (alive, no response observed yet)
State 1: Responded, on treatment (alive, confirmed response achieved)
State 2: Progressed (or off treatment, alive after progression)
State 3: Death (absorbing state)
```

This model requires specifying **explicit transition intensities** for each possible direct transition. In a 4-state model with states 0-3, there are 6 possible one-way transitions (each ordered pair (i,j) with i < j, since backward transitions like response → no response are not allowed in oncology):

$\lambda_{01}(t) = \alpha_{01} \cdot t\text{^}(\gamma_{01}-1)$ — No Response → Response (response event)
$\lambda_{02}(t) = \alpha_{02} \cdot t\text{^}(\gamma_{02}-1)$ — No Response → Progression (progression without prior response)
$\lambda_{03}(t) = \alpha_{03} \cdot t\text{^}(\gamma_{03}-1)$ — No Response → Death (death without response or progression)
$\lambda_{12}(t) = \alpha_{12} \cdot t\text{^}(\gamma_{12}-1)$ — Response → Progression (loss of response / progression after resp
$\lambda_{13}(t) = \alpha_{13} \cdot t\text{^}(\gamma_{13}-1)$ — Response → Death (death while in response)
$\lambda_{23}(t) = \alpha_{23} \cdot t\text{^}(\gamma_{23}-1)$ — Progression → Death (death after progression)

Each λ_{ij}(t) is a Weibull hazard with scale α_{ij} (arm-specific) and shape γ_{ij}. All intensities depend on arm. This generates: - **ORR** = cumulative incidence of 0→1 by t_1 (6-12 weeks) - **DOR** = sojourn time in State 1 before transition to 2 or 3 - **PFS** = min(times to 0→2, 0→3, and for responders 1→2, 1→3) - **OS** = time to State 3 from any state

Identifiability: All 6 transitions are identifiable if each is observed at least once. Rare transitions (e.g., 1→3, death while in response) may have few events. For calibration: PFS curve informs $\lambda_{02} + \lambda_{03}$; DOR curve informs $\lambda_{12} + \lambda_{13}$; OS curve provides the overall constraint on λ_{03}, λ_{13}, λ_{23}. A practical starting point is γ_{ij} = 1 (exponential transitions), introducing Weibull shapes only where supported by data.

Scenario 2: PFS → OS

```
For each patient:
1. Generate PFS event time from arm-specific Weibull: T_PFS ~ Weibull(α_arm, β)
2. Generate OS as: T_OS = T_PFS + t_post_progression
   where t_post_progression ~ Exp(λ_arm) or Weibull
   OR use a copula to link T_PFS and T_OS directly
```

5.2 Simulation Parameters

Parameter	Suggested Values	Notes
Number of treatment arms (K)	2, 3, 4	Including control
Per-arm sample size (Stage 1)	30-100 per arm (explore n=30-40 scenarios)	~15-50 events for TTE; many Phase IIs have 20-40 evaluable patients per arm
Total sample size (Stages 1+2)	300-800	Driven by primary endpoint power
ORR for control arm	0.10-0.30	Tumor type dependent
ORR for experimental arms	0.15-0.50	Vary across arms
OS hazard ratio	0.70-1.00 (vs control) — include 0.70–0.85 range for realistic solid-tumor effects	HR of 0.60 is unrealistically optimistic for most solid tumors
Correlation ORR ↔ OS	ρ = 0.3, 0.5, 0.7	Critical sensitivity parameter
Accrual rate	10-30 patients/month	Realistic for oncology
Interim analysis timing	After N_stage1 patients or E_stage1 events	Driven by short-term endpoint maturity

5.3 Decision Rules at Interim

Pick-the-Winner Rule

```
For each experimental arm j = 1,...,K:
  - Compute ORR_j (or DOR median, or PFS HR)
  - For ORR: select arm with highest ORR if ORR_j - ORR_control > δ_min
  - For DOR: select arm with longest median DOR if DOR_median_j > DOR_control × 1.3
  - For composite: rank arms by a weighted score: S_j = w₁ × Z_ORR_j + w₂ × Z_DOR_j

Futility stopping:
  - Drop arm j if ORR_j ≤ ORR_control (or if upper bound of CI excludes meaningful benefit)
  - Or use pre-specified conditional power for OS

Go/no-go for stage 2:
  - If no arm meets selection criteria → stop trial for futility
  - If exactly one arm meets criteria → select that arm
  - If multiple arms meet criteria → select the best, OR carry forward the top 2
```

Advanced Decision Rules

1. **Benefit-cost ratio approach** (Sun et al. 2020): Use an objective function that balances expected effect size against remaining sample size cost
2. **Predictive probability**: P(treatment_arm ≈ control_at_final | interim_data) for futility; P(treatment_arm > control | interim_data) for go decision
3. **Conditional power**: Probability of reaching significant OS result given interim data and assumed effect

5.4 Performance Metrics

Metric	Definition	Target
Type I error (FWER)	P(reject any true H_0)	≤ 0.025 (one-sided)
Power	P(reject H_0 for truly effective arm)	≥ 0.80 - 0.90
Probability of Correct Selection (PCS)	P(select the truly best arm at interim)	Higher is better
Probability of Futility Stop	P(stop for futility when all arms are ineffective)	≥ 0.80
Expected sample size	E[N] under null, alternative, and selected scenarios	Lower is better
Expected trial duration	E[Duration]	Lower is better
Bias in treatment effect estimate	E[β̂_selected] - β_true	Small or correctable

5.5 Sensitivity Analyses

Sensitivity Factor	Why It Matters
Correlation ρ(ORR, OS)	Selection efficiency depends critically on this
Delayed separation (non-PH)	IO agents often violate PH; impacts power and interim decision
Accrual rate variation	Slower accrual means less OS data at interim
Number of arms (K)	More arms → more selection pressure → greater bias
Interim timing	Too early: unreliable ORR/DOR; too late: minimal efficiency gain
Decision rule threshold	Stringency affects both type I error and power
Drop-the-loser vs pick-the-winner	Different operating characteristics
Carry one vs carry two arms	Trade-off between power and multiplicity adjustment
Misspecified DOR model	DOR can be longer than expected (especially with IO)

5.6 Simulation Study Implementation Plan

```
Phase 1: Null scenario
  - All arms have the same effect as control
  - Assess type I error across all configurations

Phase 2: Alternative scenarios
  - Scenario A: One effective arm, rest null
  - Scenario B: Two moderately effective arms, one null
  - Scenario C: All arms effective (dose-response)
  - Scenario D: Effective arm has weak ORR but strong OS (discordance)

Phase 3: Sensitivity
  - Vary ρ across {0.3, 0.5, 0.7}
  - Vary interim timing
  - Compare decision rules

Phase 4: Comparison with standard approach
  - Separate Phase II (pick winner) + Phase III
  - Compare sample size, duration, power
```

5.7 Software Recommendations

Tool	Purpose	Notes
R	Core simulation	Widest range of packages

Tool	Purpose	Notes
`gsDesign`	Group sequential boundaries	Lan-DeMets alpha spending, O'Brien-Fleming
`asd`	Adaptive seamless design simulation	Treatment and subgroup selection; flexible endpoint types. **Note:** `asd` was archived from CRAN as of late 2024 — verify availability before building workflow around it
`rpact`	Adaptive designs (combination test, conditional error)	**Recommended alternative to `asd`** — actively maintained, regulatory-grade implementation, supports inverse normal combination test and conditional error approaches
`simtrial`	TTE simulation with non-PH	Delayed effects, crossing hazards
`copula` / `copulaedas`	Joint modeling of endpoints	Flexible correlation structures
`SimDesign`	Structured simulation studies	Parallel computing, error handling
SAS	Alternative for regulated environments	`SEQDESIGN`, `ADAPTIVEREG`

5.8 Suggested Monte Carlo Setup

- **Simulation replications:** 5,000-10,000 per scenario (for 0.025 type I error, need ≥10,000 for reliable estimation)
- **Bootstrap confidence intervals** for operating characteristics
- **Variance reduction:** Common random numbers across scenarios to isolate design effects
- **Parallelization:** Use `parallel` or `future` packages for speed

5.9 Intern Implementation Timeline (8 Weeks)

Week 1–2: Literature foundation and first-principles implementation - Read Stallard & Todd (2003) — the pick-the-winner foundation — and implement their two-arm, two-stage design in R from scratch using efficient score statistics - Read Bauer & Posch (2004) to understand the bias mechanism - Read Jenkins et al. (2011) for the cohort-separation solution - Also read: Sun et al. (2020), Zhang & Jin (2025), Zhong et al. (2025), Wang et al. (2023), Hua et al. (2026) - Validate: reproduce key numerical results from Stallard & Todd (2003) Table I or II - Use `gsDesign` and `rpact` for the group sequential guts (do not code alpha spending from scratch)

Week 3–4: Extend to ORR/DOR setting and pilot simulation - Implement binary→binary simulation (ORR at interim, ORR at final) to validate the methodology without TTE complexity - **Pilot simulation phase:** Run a small grid (2–3 scenarios × 1,000 reps) to identify coding errors and computational bottlenecks before scaling up - Debug null scenarios first: confirm type I error = 0.025 ± MC error - Add the multi-state DGP with Weibull transitions (see §5.1)

Week 5–6: Full operating characteristics - Implement the full ORR→OS simulation with multi-state DGP and copula-based correlation - Run the complete simulation grid: null, alternatives (per-arm n=30–40, HRs 0.70–0.85) - Include safety-driven selection scenarios (§4.5) - Include non-PH scenarios (delayed separation typical of IO) - **Compare cohort-separation (Jenkins/Zhang) vs independent-increment (Hua) approach** - 5,000–10,000 reps per scenario; document Monte Carlo standard errors

Week 7: Sensitivity and robustness analyses - Sensitivity analysis for ρ(ORR, OS) across {0.3, 0.5, 0.7} — calibrate from published meta-analytic estimates (Prasad et al. 2015) rather than arbitrary values - Compare pick-a-winner vs drop-the-losers vs traditional sequential Phase 2+3 - Compare carry-one vs carry-two arms

Week 8: Drafting and publication prep - Draft methodology section with simulation results - Prepare figures (power curves, PCS curves, FWER contours over ρ and Δ) - Write clean documentation for reproducibility (save every simulation run with RNG seed, full parameters, and full results — not just summary statistics) - Build as R package: `R/` for functions, `inst/sims/` for scripts, `vignettes/` for documentation - Deliverable: R package + vignette + methodology note + 15-min presentation

Notes: - Weeks 1–2 will feel slow — reproducing published results is harder than it looks. Each bug in the correlation structure, selection rule, or combination test produces wrong results that take days to debug. This is normal. - Every simulation run must save the full state (RNG seed, parameter values, full results). You will need to debug a result six weeks later. - For ρ(ORR, OS): use published meta-analytic estimates (Prasad et al. 2015) rather than attempting patient-level empirical calibration from raw historical data, which is rarely accessible. - Check CRAN status of all R packages before building the simulation framework — `asd` was archived from CRAN as of late 2024; use `rpact` as the primary alternative.

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Appendix A: Suggested Next Steps

See §5.9 for the full 8-week implementation timeline. High-level summary:

1. **Read the key papers** in priority order (per timeline): Stallard & Todd (2003) → Bauer & Posch (2004) → Jenkins et al. (2011) → Sun et al. (2020) → Zhang & Jin (2025)
2. **Start with first-principles implementation** of Stallard & Todd's two-stage pick-the-winner, not Zhang & Jin's more complex design
3. **Run a pilot simulation phase** before scaling up to identify bugs early
4. **Extend to TTE endpoints** using the multi-state model from §5.1 with all 6 transitions specified
5. **Include sensitivity analyses** for ρ(ORR, OS), non-PH, safety-driven selection
6. **Build the simulation as an R package** from day one (`R/`, `inst/sims/`, `vignettes/`) for reproducibility

Appendix B: Notes on Papers Not Fully Reviewed

Many papers referenced in this review were initially accessed by abstract only. Since the initial review, the following full-text PDFs have been obtained (stored in `refs/`):

Paper	File
Stallard & Todd (2003)	`stallard2003-pick-winner.pdf`
Jenkins et al. (2011)	`jenkins2011-cohort-separation.pdf`
Magirr et al. (2012) — Biometrika	`magirr2012-biometrika-generalized-dunnett.pdf`
Zhang & Jin (2025)	`zhang2025-mams-phase2-3.pdf`
Zhong et al. (2025) — DTL	`zhong2025-drop-the-losers.pdf`
Hua, Wang & Luo (2026)	`hua2026-multiplicity-adaptive-selection.pdf`
Wang et al. (2023) — rank-based Dunnett	`wang2023-rank-based-dunnett.pdf`
Bauer & Posch (2004) — letter	`bauer2004-letter-schaefer-muller.pdf`

Papers still lacking full-text PDFs in `refs/` (abstract-only entries in the review):

- ☐ **Jin & Zhang (2021)** — HIGH PRIORITY: Multiple endpoint framework
- ☐ **Sun et al. (2020)** — HIGH PRIORITY: 2-in-1 design, benefit-cost ratio
- ☐ **Bretz et al. (2006)** — HIGH PRIORITY: Foundational framework
- ☐ **Friede & Stallard (2008)** — HIGH PRIORITY: Method comparison
- ☐ **Dixit et al. (2021)** — TTE + non-PH considerations
- ☐ **Broglio et al. (2024)** — MODERATE: Systematic review
- ☐ **Spivack et al. (2019)** — Dose modification scenarios
- ☐ **Zhu et al. (2024)** — Response-adaptive randomization

This literature review was compiled from publicly available abstracts via PubMed and Semantic Scholar. Full texts should be obtained through institutional access for a complete understanding.